HOTS

An organism is isolated from soil, which is a very low yielding one. How will you enhance the production activity?

The following techniques at practical genomic level help to improve the microbial strain. They are:

1. Selection of mutants
2. Recombination
3. Regulation
4. Genetic engineering
5. Protoplast fusion

6.4 Preservation of Industrially Important Micro Organisms

The selected microorganism of industrial interest must be preserved in its original form for any further use and research. There are different methods for microbial preservation. Suitable methods are selected based on the:

a. Type of micro-organism

b. Effect of the preservation method on the viability of micro-organism

c. Frequency at which the cultures are withdrawn

d. Size of the microbial population to be preserved

e. Availability of resources

f. Cost of the preservation method. Followings are some of the methods of microbial preservation:

a. Desiccation

This involves removal of water from the culture. Desiccation is used to preserve actinomycetes (a form of fungi-like bacteria) for very long period of time. The microorganisms can be preserved by desiccating on sand, silica gel, or paper strips.

b. Agar Slopes

Microorganisms are grown on agar slopes in test tubes and stored at 5 to –20 °C for six months. If the surface area for growth is covered with mineral oil the micro-organisms can be stored for one year.

c. Liquid Nitrogen

This is the most commonly used technique to store micro-organisms for a long period. Storage takes place at temperatures of less than -196 °C and even less in vapour phase. Microorganisms are made stationary and suspended in a cryoprotective agent before storing in liquid nitrogen.

d. Drying

This method is especially used for sporulating microorganisms (organisms that produce spores). They are sterilized, inoculated, and incubated to allow microbial growth, then dried at room temperature. The resultant dry soil is stored at 4° to 5 °C.

e. Lyophilization

This process is also known as freeze-drying. The microbial culture is first dried under vacuum, filled in ampoules (glass vessels) then frozen. This is a most convenient technique, since it is cheap to store and easy to ship. The disadvantage is that it is difficult to open the freeze dried ampoules; also, several subcultures have to be done to restore the original characteristics of the microorganisms.

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